Detecting Insider Threats: Weekly Account-Level Aggregation of Unsupervised Email Anomaly Scores

Maximillian McCourt

University of Edinburgh
August 2026

Executive Summary

Insider threats are harder to detect than external intrusions because insiders act within their authorised bounds, and at a firm the size of Lloyds, with over 60,000 employees, the constraint is not detection but attention. A monitoring system that flags even a small fraction of email traffic produces thousands of messages a week, and a firm needs to know which accounts warrant investigation, not which messages scored highly. Any system that is to change how a firm monitors email must therefore be judged on the workload it creates as much as the threats it surfaces.

This report develops and evaluates such a system on the CERT r6.2 synthetic insider-threat corpus, using 103,457 emails sent by 100 users over eighteen months, of which 98 messages from 13 accounts are known to be malicious. Rather than classifying emails directly, we describe each message by how far it departs from its own sender's established habits across eleven behavioural features, and score those departures with four unsupervised detectors. All models are fitted and applied in a single forward pass through time, so no email is ever scored using information generated after it was sent. Anomaly scores are then aggregated to the account level and each account assessed weekly, through two deliberately opposed statistics: one that accumulates gradual change over many weeks, and one that isolates a single extreme message.

Three findings follow. First, and most transferable, training on the full unlabelled corpus performed no worse than training on confirmed-clean data, so a firm that cannot label its email history loses nothing by using it as it stands. Second, the message-level system alone is not deployable: at the selected operating point (5% cutoff) it recovers 63% of malicious emails but presents 69 messages a week for review, at 1.3% precision. Third, the account-level layer resolves this: by reading one message per week and opening one investigation roughly every six weeks (4.1 messages a week in total), it cuts the workload seventeen-fold at 12.4% precision, opens an investigation into the Scenario 4 exfiltration before 73% of it has occurred, and catches the Scenario 3 account takeover at the first scheduled review after the event. The two statistics recovered different accounts and neither substituted for the other, so we recommend they be operated as separate review streams rather than combined into a single score.

We therefore recommend the account-level framework as a screening layer over existing email logs, requiring no labelled data and no change to what is already collected. Two limits bound this recommendation. The detector is scoped to email alone, and could attribute the account takeover to the compromised account but not identify the insider behind it; correlating these scores with logon and file-access records would close that gap and bring the two scenarios that generate no email into scope. The results also rest on 100 synthetic accounts, and ranking first of 100 is a materially weaker claim than ranking first of 60,000. Re-running the pipeline against CERT's full 4,000-user population, which requires no new data collection, is the natural next step before any deployment at Lloyds' scale.

Contents

1 Introduction **4**

2 Data Overview **4**
2.1 Dataset and Sampling . . . 4
2.2 Feature Engineering . . . 5

3 Machine Learning Methods **5**
3.1 Z-Score . . . 6
3.2 Principal Component Analysis (PCA) . . . 6
3.3 Autoencoder . . . 6
3.4 Ensemble . . . 6

4 Results **6**
4.1 Results Under the Novelty-Detection Training Regime . . . 6
4.2 Results of Models Trained with Pure-Unsupervised Regime . . . 7
4.3 Comparative Analysis of Novelty-Detection vs Pure-Unsupervised Regimes . . . 7
4.4 Model and Cutoff Selection . . . 7
4.5 Feature Importance . . . 8

5 Scenario Analysis **8**
5.1 Evaluation of Scenario 3 Emails . . . 9

6 User Analysis **9**
6.1 Accounts of Interest . . . 9
6.2 Weekly Account Statistics . . . 10
6.3 Two Review Streams . . . 10
6.4 Results . . . 10
6.5 Review Burden . . . 11

7 Discussion **12**
7.1 Main Findings . . . 12
7.2 Limitations . . . 12
7.3 Future Directions . . . 12

Appendix A Email Volume Hourly Distribution **15**

Appendix B Malicious Synthetic Scenarios **15**

Appendix C Correlation Matrix of Features **16**

Appendix D Ranking Model Performance Across All Cutoffs **16**

Appendix E Feature Importance by Scenario **16**

Appendix F Weekly Detection Record for the Accounts of Interest **17**

List of Figures

1 Detection Pipeline . . . 4
2 PCA Scree Plot . . . 6
3 Ensemble Precision-Recall Curves . . . 7
4 Distributions of Ensemble Scores by Malicious Scenario . . . 9
5 Timeline of Scenario 3 Emails by Ensemble Score . . . 9
6 CUSUM Trajectories for CDE1846 . . . 11
7 CUSUM Trajectories for CMP2946 . . . 11
8 Sending Distribution by Hour . . . 15

List of Tables

1 Malicious Scenarios of CERT r6.2 which include emails sent. . . . 4
2 Deviation Features . . . 5
3 Model Performance on Novelty Detection Training Regime . . . 7
4 Model Performance on Pure-Unsupervised Training Regime . . . 7
5 Permutation Feature Importance at the 5% Cutoff . . . 8
6 Ensemble Performance by Malicious Scenario . . . 8
7 Weekly Ranks of Accounts of Interest . . . 10
8 Detection and Burden Against the Decision Interval . . . 11
9 The five malicious scenarios of CERT r6.2 . . . 15
10 Full Correlation Matrix of Features . . . 16
11 Ranking Performance Across All Cutoffs . . . 16
12 Per-Scenario Permutation Importance . . . 16
13 Weekly Detection Record . . . 17

1 Introduction

Protecting large firms requires evaluating external attacks and internal malicious activity. Insiders act within authorised bounds, and thus their behaviour is far harder to distinguish from ordinary work than an external intrusion [4, 8]. Such cases are also very rare relative to the volume a firm the size of Lloyds produces, with over 60,000 employees [12]. This is why precision matters as much as recall: flagging even a small fraction of all emails would bury an analyst, however many threats it caught [2].

Email is a vital channel to monitor, its records carrying attachments, recipients, timestamps and content, the last of which can be represented numerically using pre-trained language models [16]. Evaluating users' deviations in these features lets us identify shifts in their sending patterns for further checking.

Machine learning models can be trained on non-malicious email to learn what looks irregular relative to a user's patterns. This assumes a labelled corpus, which a firm at Lloyds' scale does not have. We therefore also test whether training on all email, malicious cases included, is a viable alternative [9].

Finally, we address the volume such a system flags. Reviewing thousands of messages is not something a security function does; it investigates people. We therefore aggregate to the account level and assess each account weekly, opening a small number of investigations and flagging a single email each week rather than a list of messages.

This report aims to offer large firms a viable, low-overhead method of monitoring email for insider threats, maximising threats surfaced while minimising the workload needed to check them.

Figure 1 sets out the modelling pipeline in full; the sections that follow develop each stage in turn.

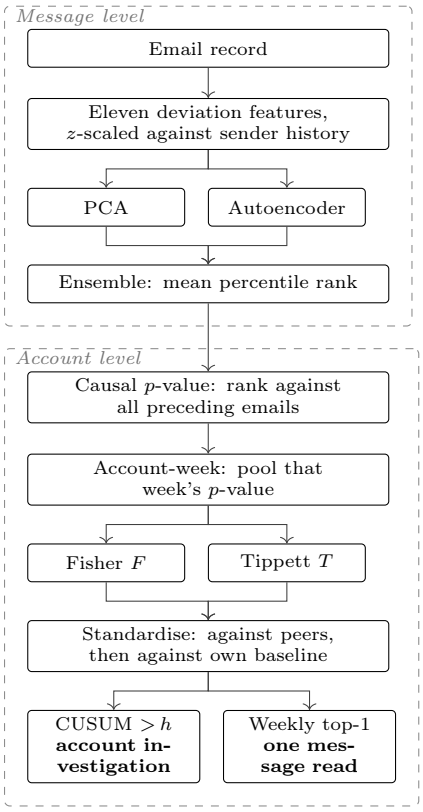


Figure 1: The detection pipeline. At message-level detection, an email is described by its deviation from its sender's own history and scored by two structural detectors whose percentile ranks are averaged. At account-level detection, that score becomes a causal p-value against all preceding emails, pooled per account-week by two opposed combination statistics, standardised against contemporaries and against the account's own baseline, and routed to the review stream matched to the threat each detects.

2 Data Overview

2.1 Dataset and Sampling

We use the CERT Insider Threat Test Dataset (r6.2), a synthetic corpus produced by Carnegie Mellon University's Software Engineering Institute [11]. It simulates the activity of 4,000 employees of a fictional organisation between December 2009 and May 2011, with an answers file labelling every injected malicious action. Its synthetic construction avoids the privacy and labelling constraints that limit real insider-threat data [7], at some cost to realism (Section 7.2).

Malicious activity spans five scenarios across four activity types: email, logon, file, and web (http) activity. Scenarios 1 and 5 involve only removable-media and web-upload activity, generating no malicious email, so fall outside this analysis; the remaining three are summarised in Table 1; all five are reproduced verbatim in Appendix B. We retain only malicious records of type email with activity Send, since viewing one is not itself malicious, leaving 98 malicious emails sent by 13 users.

Table 1: Malicious Scenarios of CERT r6.2 which include emails Sent. These are indexed, the behaviour of each synthetic scenario detailed, and the number of emails sent in each listed.

Scenario	Behaviour	Emails
2	Departing employee solicits a competitor, then exfiltrates data by thumb drive before leaving	13
3	Disgruntled system administrator keylogs his supervisor, logs in as them, and sends an alarming mass email, causing panic among the office	15
4	User accesses a colleague's machine and emails files of interest to their home address, at escalating frequency over three months	70

The full dataset is prohibitively large (∼22GB, 4,000 users). We therefore subset it to the 13 users with at least one malicious email plus 87 randomly selected users with none: 103,457 emails from 100 users. Retaining every malicious case was deliberate, since at the population base rate a sample this size would contain too few positives to evaluate. This is case–control sampling: our detectors are trained without labels on normal behaviour, so the enrichment can-

not inform them. It does raise the evaluation set's malicious base rate to 0.1%, so we treat precision figures as optimistic [17].

The accompanying LDAP files give monthly snapshots of each user's role, department and supervisor, used to construct peer-group baselines (Section 2.2). Where a record is absent, typically following departure, we carry forward the last recorded values.

2.2 Feature Engineering

Our features describe not the email itself but its deviation from the sender's ordinary behaviour, organised into eleven features across four groups (Table 2).

Table 2: The eleven deviation features. Each is expressed as a z-score against the sender's own prior distribution (Section 2.2).

Feature	Measures
Metadata	
`log_size`	Message size, log-transformed
`n_attachments`	Attachment count
`n_recipients`	Recipient count
`is_external`	Any recipient outside the domain
`external_sender`	Sender's address outside the domain
Temporal	
`is_weekend`	Sent outside the working week
`is_after_hours`	Sent outside 07:00–19:00
`log_volume_7d`	Emails sent in the preceding week, log-transformed
Contact network	
`contact_surprise`†	How unexpected the recipient set is
`frac_novel_recipients`†	Proportion of recipients never previously contacted
Semantic	
`semantic`‡	Cosine distance from the sender's topic centroid

Unmarked features require 30 prior emails, or the peer-group baseline that replaces it where the sender's history is shorter. † 10 prior recipient slots. ‡ 5 prior messages.

Two require definition beyond their description there. `contact_surprise` measures how unexpected the recipient set is given the sender's history: writing $c_u(r)$ for the number of prior emails from u to r, and $m_u = \max_r c_u(r)$ for their most-contacted recipient, an email to recipient set R scores

$$\texttt{contact_surprise} = 1 - \frac{1}{|R|} \sum_{r \in R} \frac{c_u(r)}{m_u},$$

so values near zero indicate close contacts and values near one, people never contacted. The `semantic` feature embeds each message using the pretrained sentence transformer `all-MiniLM-L6-v2` [16, 22], giving a 384-dimensional representation, and falls back to the preceding month's organisation-wide centroid where the sender has fewer than five prior messages. Message size and email-volume are log-transformed before standardisation to correct heavy right-skew, and the after-hours threshold is set from the population's sending distribution (1.02% of traffic falls outside it, see Appendix A).

Each deviation feature is z-scaled against its distribution across the sender's prior emails; for the network and semantic features this is a second standardisation, the underlying quantity being already a novelty measure. Conditioning on prior activity keeps the construction causal, and personal standard deviations let one model train across all users rather than one per user. Where a user's prior emails vary little in a feature its standard deviation approaches zero, so we clip all deviations to $[-10, 10]$; fewer than 0.05% of emails reach this bound, leaving score ordering almost unaffected.

Baselines are assigned hierarchically for users with limited history: those with at least 30 prior emails use their own expanding history; failing that, we descend through broader LDAP peer groups (role, department, business unit) to an organisation-wide baseline. Peer and organisation-wide baselines use the preceding calendar month only, so they track current behaviour rather than averaging over the full observation window.

4,081 emails (3.9%) lack a complete feature vector and are excluded from scoring rather than imputed, since a substituted value would assert typicality where none exists; none are malicious.[1]

Collinearity. The strongest correlation among the eleven deviation features is 0.89, between `is_external` and `external_sender`, followed by 0.72 between `log_size` and `n_attachments` (see Appendix C). We retain both pairs: the former record different things, a recipient's domain versus the sender's own, and the latter can diverge too, since a message may be large through body text alone or through many small attachments. The remaining features correlate at 0.55 or below, and most at under 0.16.

3 Machine Learning Methods

All models are fitted and scored in a single forward pass through the corpus ordered by time, so no email is scored using information generated after it was sent: scoring begins once 1,000 emails have accumulated, with a refit every further 1,000. Refitting periodically rather than per-email cuts computation while keeping the model applied to any email always fitted on a strict subset of its past.

Models are fitted only on non-malicious emails, the novelty-detection setting; Section 4 tests the alternative of training on the full, unlabelled corpus.

[1] 2,569 emails (2.4%) have no baseline, those of the first month of the observation window. Two further features are individually missing: `contact_surprise` for 1,512 more emails (fewer than ten prior recipient slots) and `semantic` for 375 more (fewer than five prior messages). The union gives the 4,081 excluded.

3.1 Z-Score

Our simplest detector treats each feature independently, scoring departure from the origin, an email exactly typical for its sender, as the root mean square of the deviation vector [24]:

$$s_i = \sqrt{\frac{1}{p}\sum_{j=1}^{p} x_{ij}^2}, \tag{1}$$

where i indexes emails, j indexes features and $p = 11$.

This is the special case of a multivariate distance in which the covariance matrix is assumed to be the identity, so a jointly implausible but individually unremarkable combination draws no more attention than an isolated extreme value. This assumption is not satisfied here, correlation of up to 0.89 persisting among the retained features (Appendix C), so we report the z-score as a reference point rather than combine it with the models that follow (Section 3.4).

3.2 Principal Component Analysis (PCA)

Principal component analysis relaxes the independence assumption by learning normal email's dominant correlation structure. On each refit it is fitted to the training window, retaining $k = 5$ leading components; each email is then projected into this subspace and reconstructed, scoring the mean squared reconstruction error [19]:

$$s_i = \frac{1}{p}\left\| x_i - \hat{x}_i \right\|^2. \tag{2}$$

Emails reconstruct poorly when their feature combinations do not fit this learned structure; PCA is, however, blind to anomalies along the high-variance directions it retains, since reconstruction error lives only in the residual subspace. We fix k rather than the variance-explained proportion, which would let k drift between refits as the training window's covariance evolves, undermining comparison of scores across time.

Figure 2 provides a scree plot on non-evaluated data, used for determining the optimal k.

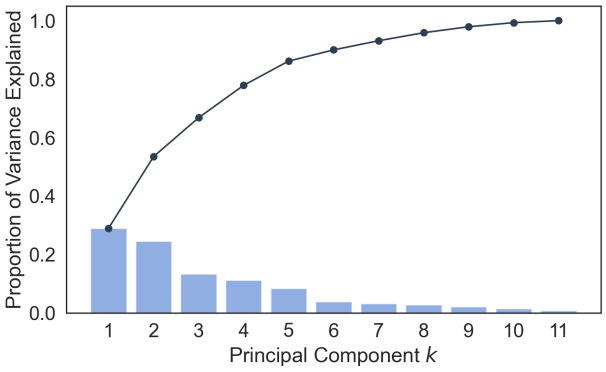


Figure 2: PCA Scree Plot, used for validating number of principal components k used for reconstructing data by proportion of variance explained. This is computed on the first 1,000 emails (those excluded from scoring due to the burn-in period), and thus does not indicate any model leakage.

The sharpest break in cumulative variance occurs between the fifth and sixth components (Figure 2), the leading five explaining 86.1% and the rest under 4% each. We adopt $k = 5$, chosen visually from training data alone and uninformed by labels or the evaluation period.

3.3 Autoencoder

To capture nonlinear structure we introduce a neural autoencoder with an $11 \to 8 \to 4 \to 8 \to 11$ architecture: a rectified linear activation after the first layer of each half, with a linear four-dimensional bottleneck and a linear output [18]. As with PCA, the anomaly score is the mean squared reconstruction error, keeping both on a common scale; but its learned manifold is curved, capturing feature interactions PCA can only approximate linearly.

Each refit trains for 10 epochs using Adam (learning rate 0.005, batch size 64), warm-starting from the previous fit so the network retains what it has learned as the training window expands. Hyperparameters were set a priori rather than tuned (Section 7.2).

3.4 Ensemble

Finally we combine the two structural detectors: since PCA and autoencoder reconstruction errors are not calibrated against one another, we convert each model's scores to percentile ranks and average them with equal weight, a method invariant to any monotone transformation and symmetric between the two models.

The z-score is structurally redundant here: since the deviation features are approximately centred, Equation 1 is the limiting case of Equation 2 with no subspace retained, so including it would weight the same overall departure twice and add nothing. Equal weighting is imposed a priori, following established practice in unsupervised ensembles for insider threat detection [10]: no component weight is informed by the label against which it is subsequently evaluated.

4 Results

This section reports detection performance under both the novelty-detection and pure-unsupervised regimes, training on only non-malicious emails versus all emails, and selects the model, training regime and cutoff carried forward for the remainder of this report.

Every recall figure below carries a 95% Wilson confidence interval on the 98 malicious emails [23]; precision is a point estimate only, since its denominator is fixed by the review budget rather than sampled. These intervals are a lower bound on the true uncertainty, since the 98 emails arise from a small number of injected incidents rather than 98 independent events (Section 7.2).

4.1 Results Under the Novelty-Detection Training Regime

Table 3 shows the model results under the novelty-detection training regime. The z-score needs no training data, so we report it once, as a reference point (Section 3).

Table 3: Detection performance by score cutoff (n=98 malicious emails), with 95% Wilson intervals on recall [23]. The 1%, 5% and 10% cutoffs correspond to 984, 4,919 and 9,838 of the 98,376 scored emails respectively.

Model	Cutoff	Caught	Recall (95% CI)	Precision
Z-score	1%	5/98	5.1% (2.2, 11.4)	0.5%
	5%	10/98	10.2% (5.6, 17.8)	0.2%
	10%	37/98	37.8% (28.8, 47.6)	0.4%
PCA	1%	10/98	10.2% (5.6, 17.8)	1.0%
	5%	56/98	57.1% (47.3, 66.5)	1.1%
	10%	72/98	73.5% (64.0, 81.2)	0.7%
Autoencoder	1%	22/98	22.4% (15.3, 31.7)	2.2%
	5%	62/98	63.3% (53.4, 72.1)	1.3%
	10%	82/98	83.7% (75.1, 89.7)	0.8%
Ensemble	1%	24/98	24.5% (17.0, 33.9)	2.4%
	5%	61/98	62.2% (52.4, 71.2)	1.2%
	10%	80/98	81.6% (72.8, 88.1)	0.8%

Z-score performs worst throughout on recall, its confidence interval at 10% (28.8%, 47.6%) sitting entirely below the structural models'. PCA is similarly weak at 1% (10.2%) but improves sharply thereafter, suggesting most malicious emails sit between the 95th and 99th percentile of its ranking. The autoencoder outperforms PCA substantially at 1% (22.4%), with barely-overlapping intervals, and more modestly after; the ensemble's intervals largely overlap the autoencoder's, never differing by more than one or two emails. Both also lead PCA on precision at every cutoff.

4.2 Results of Models Trained with Pure-Unsupervised Regime

Reflecting the constraint facing a firm the size of Lloyds, which cannot label its full email corpus, we repeat the above with all emails included in training (model results shown in Table 4).

Table 4: Detection performance trained on all data, contamination included (n=98 malicious emails), with 95% Wilson intervals on recall [23]. The 1%, 5% and 10% cutoffs correspond to 984, 4,919 and 9,838 of the 98,376 scored emails respectively. Z-score is omitted, as its computation does not depend on any training window (Section 3).

Model	Cutoff	Caught	Recall (95% CI)	Precision
PCA	1%	10/98	10.2% (5.6, 17.8)	1.0%
	5%	55/98	56.1% (46.3, 65.5)	1.1%
	10%	72/98	73.5% (64.0, 81.2)	0.7%
Autoencoder	1%	23/98	23.5% (16.2, 32.8)	2.3%
	5%	63/98	64.3% (54.4, 73.1)	1.3%
	10%	81/98	82.7% (74.0, 88.9)	0.8%
Ensemble	1%	24/98	24.5% (17.0, 33.9)	2.4%
	5%	62/98	63.3% (53.4, 72.1)	1.3%
	10%	80/98	81.6% (72.8, 88.1)	0.8%

4.3 Comparative Analysis of Novelty-Detection vs Pure-Unsupervised Regimes

All three models are essentially unaffected by training regime at every cutoff. The largest difference anywhere between the two tables is a single malicious email, scattered inconsistently across models and cutoffs, and every point estimate under one regime falls comfortably inside the interval of the other. Contamination at a base rate of 0.1% is, on this evidence, immaterial to any of the detectors considered. Appendix D confirms this without reference to any cutoff: no model differs by more than 0.0002 on PR-AUC or AP@10%, or 0.002 on ROC-AUC, between regimes.

Figure 3 confirms this. The grey dashed line marks the precision of flagging emails at random at our 0.1% base rate; both regimes sit at least an order of magnitude above it, converging only as recall approaches 100% and the model exhausts its ability to discriminate malicious from routine activity. The two curves track one another throughout, their 5% operating points differing by roughly one percentage point of recall. Since labelled data is unrealistic at Lloyds' scale, we adopt the pure-unsupervised regime for the remainder of this report, a materially useful finding since a bank without the means to label its email traffic need not forgo this style of detection.

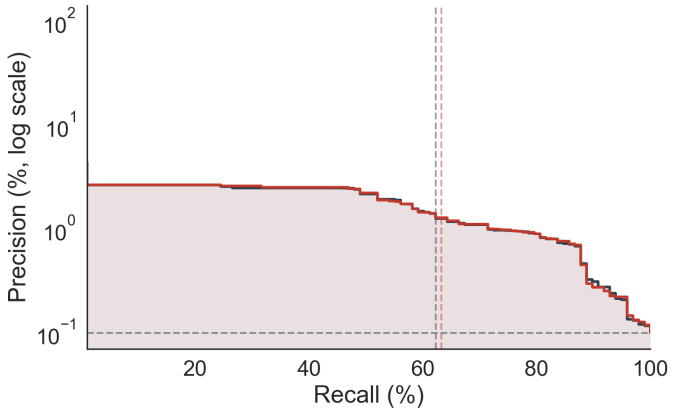


Figure 3: Precision–recall curves for the ensemble model under both training regimes, with precision on a log scale. Dashed vertical lines mark the recall each regime attains at the 5% cutoff adopted in Section 4.4 (red, pure-unsupervised; blue, novelty-detection). The grey dashed line gives the precision expected from flagging emails at random at the population base rate.

4.4 Model and Cutoff Selection

We now select between the autoencoder and the ensemble within the pure-unsupervised regime. The difference between them is not significant: the autoencoder catches one more malicious email at 5% and 10%, the ensemble one more at 1%, each difference well within the 95% confidence interval of the corresponding recall estimate. We therefore cannot select on measured performance.

Selection instead rests on a criterion fixed independently of this comparison's outcome: the ensemble requires a linear and a nonlinear characterisation of normal behaviour to agree before an email scores highly, suppressing false posi-

tives neither model corroborates alone. This makes the ensemble the more conservative choice for deployment, where an unopposed false positive costs an analyst's time regardless of which model produced it [1]. We therefore select the ensemble, its component weights already fixed a priori (Section 3.4).

We adopt the 5% cutoff: recall more than doubles versus 1% (63.3% vs. 24.5%) at roughly half the precision (1.3% vs. 2.4%), and given the rarity of these emails we prefer a cutoff capturing the majority. The 10% cutoff gains further recall (81.6%, non-overlapping interval) far less productively: the second 4,919 reviews recover only 18 malicious emails, a marginal precision of 0.37% against 1.26% for the first. 5% therefore balances recall against workload, though 4,919 messages is no reviewable queue; Section 6 moves the unit of decision from messages to accounts.

4.5 Feature Importance

We ask which of the eleven deviation features the ensemble relies on using permutation importance [3]: each is shuffled across the corpus and every email re-scored using the models already fitted, the fall in recall at the 5% cutoff measuring dependence on it. Because no model is refitted, this isolates the feature's own contribution rather than confounding it with a change in what the model has learned; we average over five shuffles and report results in Table 5.

Table 5: Recall lost when each feature is shuffled, in percentage points, against a baseline recall of 63.3% for the ensemble trained on all data and operated at the 5% cutoff. Intervals are 95% t-intervals across five shuffles, and so describe variability between permutations rather than sampling uncertainty in the 98 malicious emails.

Feature	Recall lost (pp)	95% CI
`log_volume_7d`	38.4	(35.5, 41.3)
`external_sender`	27.8	(22.7, 32.8)
`is_external`	20.4	(17.7, 23.1)
`contact_surprise`	17.6	(15.9, 19.2)
`n_attachments`	11.6	(9.9, 13.3)
`frac_novel_recipients`	10.0	(8.9, 11.1)
`log_size`	7.4	(5.5, 9.2)
`n_recipients`	6.7	(5.6, 7.9)
`semantic`	2.9	(1.8, 3.9)
`is_after_hours`	−0.2	(−0.8, 0.4)
`is_weekend`	−2.5	(−3.6, −1.3)

The ensemble's detection rests overwhelmingly on four features: shuffling `log_volume_7d` alone costs 38.4 percentage points of recall against a 63.3% baseline, and `external_sender`, `is_external` and `contact_surprise` each cost between 17 and 28 points. These describe a change in how much a user sends and to whom, consistent with Table 1: two of the three email-generating scenarios are exfiltration to an address the sender does not normally write to, and the third an unusual volume of mail from a compromised account. These are two distinct mechanisms rather than one, and the ordering here averages them: Appendix E separates the contributions by scenario and shows `log_volume_7d` is worth nothing to Scenario 2, while `contact_surprise`, which drives Scenario 2, contributes under a fifth of Scenario 4's detection.

The remaining features contribute far less. The semantic feature is worth 2.9 points, ninth of eleven though not insignificant at 95% confidence; we attribute this to CERT's synthetic message content and treat it as a dataset rather than an approach limitation (Section 7.2). The two temporal flags contribute nothing measurable, and `is_weekend` is mildly negative, its interval strictly below zero: the model recalls slightly more when it is randomised, suggesting weekend sending is itself a source of anomaly among non-malicious users. Two caveats bound this: permutation importance measures reliance on a feature, not how a model retrained without it would perform; and shuffling breaks correlated features' relationships too, so where `is_external` and `external_sender` correlate at 0.89 (Appendix C), each survivor partially substitutes for the one shuffled, understating both.

5 Scenario Analysis

Only Scenarios 2, 3 and 4 generate email traffic (13, 15 and 70 malicious emails respectively); Table 6 breaks ensemble performance down by scenario.

Table 6: Ensemble model performance broken down by malicious scenario (trained on all data), with 95% Wilson intervals on recall [23]. The 1%, 5% and 10% cutoffs correspond to 984, 4,919 and 9,838 flagged emails respectively. Precision is omitted here, as its denominator is the full flagged set rather than the scenario, making it a rescaling of recall by scenario size rather than a quantity in its own right.

Scenario	Cutoff	Caught	Recall (95% CI)
Scenario 2	1%	3/13	23.1% (8.2, 50.3)
	5%	13/13	100.0% (77.2, 100.0)
	10%	13/13	100.0% (77.2, 100.0)
Scenario 3	1%	1/15	6.7% (1.2, 29.8)
	5%	1/15	6.7% (1.2, 29.8)
	10%	3/15	20.0% (7.0, 45.2)
Scenario 4	1%	20/70	28.6% (19.3, 40.1)
	5%	48/70	68.6% (57.0, 78.2)
	10%	64/70	91.4% (82.5, 96.0)

At the 5% operational cutoff the model isolates Scenario 2 completely and catches the majority of Scenario 4 (68.6% recall), while Scenario 3 is detected remarkably poorly; despite wide intervals at these scenario sizes, Scenario 3 and Scenario 2's intervals do not overlap at the 5% or 10% cutoff. Figure 4 shows why: Scenarios 2 and 4 peak heavily above the 5% cutoff, while Scenario 3's distribution heavily overlaps non-malicious traffic, with no distinct anomalous peak.

Scenario 2's malicious emails, soliciting employment from a competitor, are addressed to an unfamiliar external recipient, and the model recovers them almost entirely on that basis: shuffling `contact_surprise` alone costs 64.6 of its 100 points of recall, with `external_sender`

and `is_external` costing 35.4 and 20.0 respectively (Appendix E). Scenario 4 escalates in frequency over its three-month window and is recovered on a different basis: `log_volume_7d`, worth nothing at all to Scenario 2, costs 54.6 of its 68.6 points, with early, lower-volume instances blending into the sender's baseline undetected. The corpus-wide ranking of Table 5 averages these two mechanisms, weighted towards Scenario 4 by its 70 positives.

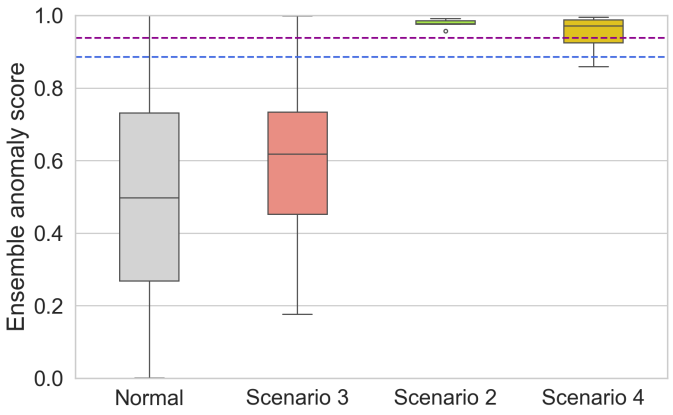


Figure 4: Box-plot distributions of ensemble anomaly scores by scenario. Non-malicious emails are labelled as 'Normal'. Boxes give the median and interquartile range. The purple and blue dashed lines show the 5% and 10% cutoffs, respectively.

Scenario 3 involves a system administrator compromising a supervisor's account to send a disruptive mass email; the takeover email is in fact the best-detected email in the scenario, so the shortfall lies elsewhere.

5.1 Evaluation of Scenario 3 Emails

We cross-referenced the 15 Scenario 3 emails against the LDAP registry to locate this shortfall. All fall within a four-day window, 12–16 August 2010 (Figure 5), and originate from eleven users in the Administration unit of the Security department: one Manager account (HIS1706), one ITAdmin insider account (PLJ1771), and nine subordinate ITAdmin accounts, most reporting directly to that manager.

The takeover email, sent from the Manager's account on 13 August, is the highest-scoring in the scenario-3 set (score = 0.9992), consistent with a disruptive mass communication. The remaining fourteen emails score substantially lower: four passed the day before between the Manager's account and the Scenario 3 insider, two each way, and ten came from nine subordinate accounts three days later, on 16 August. This ordering argues against reading the later emails as preparation for the takeover: whatever they represent, they follow the malicious incident rather than precede it.

We could not establish from the email records alone whether these ten later emails form part of the same injected incident as the account takeover or a separate, temporally adjacent one; we treat this as an open, checkable question against the dataset's ground truth rather than speculate. What is clear is the model's behaviour: its recall rests entirely on the single email in which a compromised, authenticated account sends content unlike anything its true owner has sent before. No other Scenario 3 email is flagged at the 5% cutoff, and only two further are recovered by relaxing to 10%.

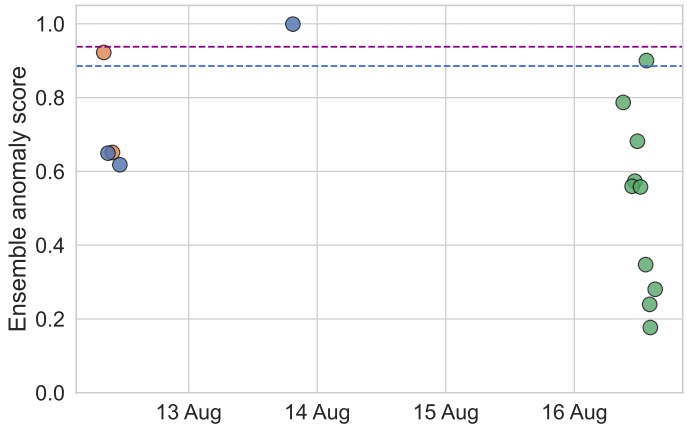


Figure 5: Temporal progression of the Scenario 3 attack cluster (August 12–16, 2010). The plot highlights the ensemble anomaly scores of the emails, distinguished by the sender's involvement. Blue dots are Scenario 3 emails of `HIS1706`, the compromised supervisor, pink are those of the insider `PLJ1771`, and green are other ITAdmin subordinates in the Administration unit. The purple and blue dashed lines indicate the 5% and 10% operational cutoff, respectively.

This is a genuine limitation rather than a tuning problem: a purely email-based detector cannot resolve behaviour leaving no trace in email content, timing or recipients, and closing the gap would require correlating its output with authentication records (Section 7.3). It also motivates a shift from scoring emails individually to characterising behaviour at the user level over time.

6 User Analysis

The ensemble scores the account-takeover email correctly, ranking it fourth-most anomalous in the corpus, yet it is flagged alongside 4,918 others: at the 5% cutoff an analyst reviews 4,919 messages over the 71-week window, some 69 per week, to recover 62 malicious emails, with no view of whether one account is responsible for many flags. We therefore change the unit of review from the message to the account, assessed weekly, retaining the ensemble, the pure-unsupervised regime and the 5% cutoff.

6.1 Accounts of Interest

Email labels do not transfer directly to accounts. Thirteen accounts sent at least one malicious email, but nine are the subordinate `ITAdmin` accounts of Section 5.1, whose intent we could not establish, and are excluded. Three appear in `insiders.csv`, the dataset's master file of true positives [11], as the actors behind Scenarios 2, 3 and 4. A fourth, `HIS1706`, is the compromised supervisor: not an insider, but the account a detector should attribute the activity to.

`PLJ1771`, the Scenario 3 insider, warrants comment: its two malicious emails are 97 minutes apart, and the disruptive mass communication was issued not from this account

but the one it compromised, so we treat `HIS1706` as the account of interest for this scenario.

6.2 Weekly Account Statistics

Each account is assessed every Sunday on the emails it sent that week. Only account-weeks with at least three emails are scored (5,879, 87.6%), totalling 97,123 emails, with a median of 16 emails per account-week and 81 accounts assessed per week. Of these, 5,079 account-weeks are rankable; the remaining 800 are scored but never enter a ranking, lacking either a cross-sectional pool of ten contemporaries or the eight prior weeks needed for a personal baseline.

Writing $\mathcal{H}_i$ for the ensemble scores of all emails preceding email i,

$$p_i = \frac{1 + |\{\, j \in \mathcal{H}_i : s_j \geq s_i \,\}|}{1 + |\mathcal{H}_i|}, \tag{3}$$

gives the proportion of prior emails at least as anomalous as this one [15]; as with Section 2.2's deviation features this conditions only on the past, and under the assumption that an account behaves like the corpus, p_i is uniform on $(0, 1]$.

For an account sending n emails in a week we combine these p-values in two ways, each targeting a different insider method:

$$F = -2\sum_{i=1}^{n} \log p_i \;\sim\; \chi^2_{2n}, \tag{4}$$

$$T = 1 - \Big(1 - \min_i p_i\Big)^n. \tag{5}$$

Equation 4 is Fisher's combination [6]. It multiplies every message's p-value together, so an account whose week contains a dozen mildly unusual emails scores highly even though no single message would attract attention on its own. This is the signature of an insider steadily changing how they work, exfiltrating a few files at a time.

Equation 5 is the Tippett statistic under Šidák correction [20, 21]. It discards all but the most extreme message and asks whether that message is more extreme than n attempts would ordinarily produce, the correction preventing high volume alone from scoring an account highly. This is the signature of a single illegitimate act on an otherwise normal account, like Scenario 3's compromised credentials. Both are carried forward as $-\log_{10}$ of the resulting p-value, larger values indicating greater anomaly.

Standardisation. Each statistic is adjusted twice. Because the ensemble refits periodically (Section 3) its score distribution is non-stationary and the p_i only approximately uniform, so a refit shifting scores upward inflates every account's statistics at once. We first convert each to a z-score against the other accounts assessed that week,

$$z_{u,t} = \frac{S_{u,t} - \mu_t}{\sigma_t}, \tag{6}$$

cancelling corpus-wide movement. We then subtract the account's own expanding mean over preceding weeks, $\tilde{z}_{u,t} = z_{u,t} - \bar{z}_{u,<t}$, following Section 2.2, to account for users who are typically unusual. We centre but do not rescale: an expanding baseline estimates dispersion from few observations, and a small denominator would favour accounts whose behaviour varies least.

6.3 Two Review Streams

The two statistics describe different threats and warrant different units of review.

Sustained shift. A gradual exfiltration campaign raises the proportion of unusual messages over many weeks without any single week being remarkable, so we accumulate the centred Fisher statistic using a cumulative sum (CUSUM) [14],

$$C_{u,t} = \max\left(0,\; C_{u,t-1} + \tilde{z}_{u,t} - k\right), \tag{7}$$

opening an account investigation when $C_{u,t} > h$ and resetting thereafter. The reference value $k = 0.5$, half the shift the chart is tuned to detect, targets a one-standard-deviation shift in an account's weekly Fisher statistic [13]. A chronically unusual account therefore does not accumulate toward an alert. Where too few emails are sent in a week to score, the statistic carries forward unchanged.

Point anomaly. A compromised account may send a single message unlike anything its owner has sent before, which no accumulation will recover, so each week we flag the most extreme message of the account with the largest centred Tippett statistic.

We fix $h = 6$, yielding eleven account investigations over the 71-week window, and report performance across $h \in \{4, 6, 8, 10\}$ in Table 8; the threshold is a review budget set by analyst capacity, not a statistical criterion.

6.4 Results

Table 7 reports the in-window weeks in which each account of interest ranked highly on either statistic. Appendix F provides the full record.

Table 7: Rank among the accounts assessed that week, for the in-window weeks in which the Fisher statistic placed the account in the leading three. `PLJ1771` is omitted, as it reached rank 41 on Fisher and 37 on Tippett in its single in-window week. `HIS1706` is included for its occurrence in which the Tippett statistic placed it in the leading three.

Account	Week ending	Emails	Malicious	Fisher	Tippett
`CDE1846`	27 Mar 2011	31	13	1	4
	03 Apr 2011	15	6	2	7
	17 Apr 2011	40	22	1	2
	01 May 2011	33	17	1	12
`CMP2946`	27 Feb 2011	34	6	1	5
`HIS1706`	15 Aug 2010	32	3	53	1

These weekly ranks track malicious volume directly. Of `CDE1846`'s ten in-window weeks, the five carrying no more than three malicious emails rank between 12th and 29th, the two carrying six rank second and ninth, and the three car-

rying thirteen or more rank first, the detector responding to the escalation in proportion to its size, as a sustained-shift detector should (Appendix F).

HIS1706 ranks 53rd of 83 on Fisher but first on Tippett in the same week, while CDE1846 is never top-ranked on Tippett within its window. Neither account is recoverable by the stream that recovers the other. Averaging the two statistics does not resolve this: it demotes HIS1706 from first to third, so the single message read that week would not have been its, and a cumulative sum on the averaged statistic never alarms on the account at all (Appendix F).

Figures 6 and 7 show why. CDE1846 accumulates from late March, crossing h on 27 March at $C = 7.75$, resetting, rebuilding over the following fortnight and crossing again on 17 April at $C = 10.61$, recovering twice because the underlying behaviour recurs. CMP2946 rises to $C = 4.58$ in the week ending 27 February on a single week ranked first on Fisher, then decays: the evidence is real but does not sustain. The chart is built to distinguish a campaign from a single unusual week, and does so here at the cost of one insider detection.

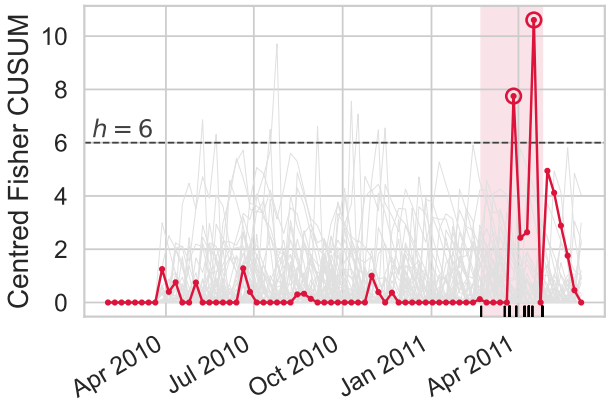


Figure 6: Cumulative sum of the centred Fisher statistic (Equation 7) for CDE1846, with the remaining accounts in grey. Circled points on the CUSUM are where CDE1846 passes the $h = 6$ threshold (marked by the dashed line). The red shaded area is the incident window, and ticks are the individual malicious messages.

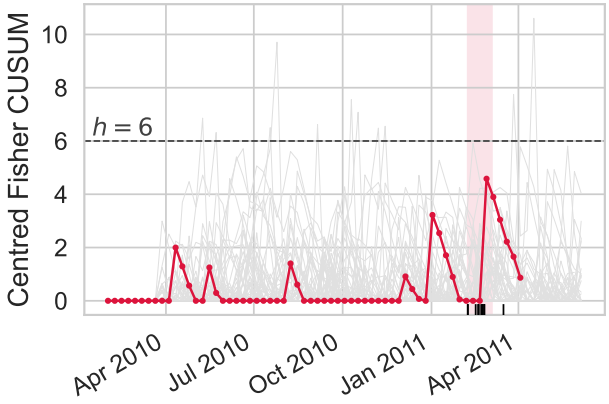


Figure 7: Cumulative sum of the centred Fisher statistic (Equation 7) for CMP2946, with the remaining accounts in grey. Shading marks the incident window, the dashed line the decision interval $h = 6$, and ticks the individual malicious messages.

6.5 Review Burden

At $h = 6$ the system opens eleven account investigations across the 71-week window and reads the most extreme message in each of the 63 weeks that produce a Tippett ranking: 290 messages in total, 4.1 per week, of which 36 are malicious, a precision of 12.4%. Against the message-level system of Section 4.4, at 69 messages per week and 1.3% precision, this reads one-seventeenth as many at roughly ten times the precision.

Email recall is not the standard by which this analysis should be judged. A message-level detector recovering 62 of the 98 malicious emails produces 62 flags within a queue of 4,919 and never names the account responsible, while an account-level detector produces a single alarm, after which further messages are activity being watched rather than being missed. The operative question is therefore when an investigation opens relative to the incident timeline.

CDE1846 alarms in the week ending 27 March 2011, only 19 of its 70 malicious emails sent at this point, so an investigation opens before 73% of the Scenario 4 campaign. HIS1706 alarms in the week ending 15 August 2010, the first review following the takeover. CMP2946 is not recovered at this threshold, and PLJ1771 is not recoverable in principle (Section 6.1).

Table 8: Detection and review burden as the decision interval h varies. In-window alarms are reported for each account of interest; HIS1706 is recovered by the Tippett stream, which does not depend on h. Emails read combines the account investigations with at most one message per week from the Tippett stream. PLJ1771 is omitted, alarming in none of its in-window weeks at any h.

h	Invest.	Emails/wk	Mal.	Prec.	CDE1846	CMP2946	HIS1706
4	37	12.8	59	6.5%	3/10	1/4	1/1
6	11	4.1	36	12.4%	2/10	0/4	1/1
8	3	2.0	29	20.6%	2/10	0/4	1/1
10	2	1.6	24	21.6%	2/10	0/4	1/1

Table 8 bounds the threshold choice: $h = 4$ recovers CMP2946 and a third CDE1846 week at three times the investigations and half the precision, while detection of CDE1846 and HIS1706 is unchanged across $h \in \{6, 8, 10\}$, so neither finding rests on the operating point. Raising h to 10 leaves two investigations across the entire sixteen-month window, both CDE1846: at that budget every account investigated is the insider.

These precision figures should be read as optimistic, since the case–control enrichment of Section 2 raises the account-level base rate well above its population value; remaining limitations are discussed in Section 7.2. Subject to that, the aggregation delivers what a security function can use: an analyst reading one message a week and opening one investigation every six weeks contains the Scenario 4 exfiltration before its campaign concludes, and catches the Scenario 3 takeover at the first review after it, each through the stream built for it.

7 Discussion

7.1 Main Findings

The central finding of this report is that machine learning can score emails for anomalous signals without a labelled corpus, and that these scores aggregate into a weekly account-level framework capturing both long-term exfiltration and single alarming messages within a manageable workload.

The most transferable of these results is that this holds under contamination: training on the full, unlabelled corpus performs no worse than training on confirmed-clean data (Section 4.3), so a firm that cannot label its email history loses nothing by using it anyway.

The second is methodological: the two account-level streams recover different accounts. `CDE1846` is caught only by the Fisher-based CUSUM, and `HIS1706` only by the Tippett stream (Section 6.4), so retaining them as separate metrics rather than averaging them into one is not a design preference but a requirement of the evidence (Appendix F). This closes the gap Section 5.1 identified: an email-only detector recognised the Scenario 3 takeover message, fourth-most anomalous in the corpus, without linking it to an account, and the Tippett stream now does exactly that. Le and Zincir-Heywood [10] report unsupervised ensembles on CERT across all four activity types at user-day granularity; our email-only, user-week formulation is not directly comparable, but both find that combining dissimilar detectors outperforms either alone under unlabelled training.

The message-level system alone cannot notice that one account is responsible for many flags, having no aggregation step above the individual email. Weekly, account-level review replaces the roughly 70 messages a week it would require with eleven investigations and one flagged email per week (Section 6), at a fraction of the workload and an increased precision.

7.2 Limitations

This analysis rests on a subset of CERT r6.2 in which three users are labelled insiders and one is compromised; in reality this proportion is likely far smaller, threats being exceedingly rare at a firm like Lloyds, so we are cautious in any conclusions drawn from the feature importance of Section 4.5. The data are also synthetic, and unlikely to capture the complexity of real sending habits across a large firm's departments: the 2.9-point recall cost of the semantic feature is likely an underestimate, since CERT's message content is generated from source text by topic rather than written for a communicative purpose [7].

A related limitation is dataset size: 100 accounts against Lloyds' roughly 60,000. Ranking first of 100 is a fundamentally different claim from ranking first of 60,000, and both the top-1 Tippett stream and CUSUM's investigation count would need to scale, roughly linearly for the latter, with h raised accordingly to keep review volume manageable.

This analysis is also scoped to email alone: Scenarios 1 and 5, which generate none, fall entirely outside it, and our detector could flag `HIS1706` but not reveal `PLJ1771`'s logon behind it (Section 5.1). Insider threats in reality span takeovers, file exfiltration and online activity together, and a deployed detector should draw on all of them rather than leaving an analyst to infer the rest.

All confidence intervals likely understate true uncertainty, since the 98 malicious emails arise from only three incidents and so are unlikely to be independent (Section 4). The autoencoder's hyperparameters were also set a priori rather than tuned (Section 3.3); a sensitivity analysis over bottleneck dimension and epoch count remains open work.

7.3 Future Directions

Cost-weighted evaluation would let an analyst weight scenarios by counterfactual detection probability, so a slow-drip campaign scores more highly than an incident, like an alarming mass email, that recipients would report regardless.

The highest-value extension is multi-modal fusion: correlating the email model's output with logon and file-access records directly. This would close the Scenario 3 gap of Section 5.1, where account misuse leaves no trace in email, and bring Scenarios 1 and 5 into scope.

The Fisher stream also pools every email a user sends in a week; an account mixing malicious and ordinary activity dilutes its own score; higher criticism for sparse mixtures [5] or the truncated product method [25] could target this directly, closing an avenue by which a deliberate insider might evade the cumulative statistic.

Finally, testing how the ranking holds at scale would establish these results as actionable rather than theoretical. CERT r6.2's full population of 4,000 users (Section 2.1), set aside here for tractability, is a natural test: rerunning the pipeline and checking whether investigation counts and stream disjointness hold at that scale needs no new data collection.

References

[1] Charu C. Aggarwal. "Outlier Ensembles: Position Paper". In: *ACM SIGKDD Explorations Newsletter* 14.2 (2012), pp. 49–58. DOI: 10.1145/2481244.2481252.

[2] Bushra A. Alahmadi, Louise Axon, and Ivan Martinovic. "99% False Positives: A Qualitative Study of SOC Analysts' Perspectives on Security Alarms". In: *31st USENIX Security Symposium (USENIX Security 22)*. USENIX Association, 2022, pp. 2783–2800.

[3] Leo Breiman. "Random Forests". In: *Machine Learning* 45.1 (2001), pp. 5–32. DOI: 10.1023/A:1010933404324.

[4] Dawn Cappelli, Andrew Moore, and Randall Trzeciak. *The CERT Guide to Insider Threats: How to Prevent, Detect, and Respond to Information Technology Crimes (Theft, Sabotage, Fraud)*. Jan. 2012.

[5] David Donoho and Jiashun Jin. "Higher Criticism for Detecting Sparse Heterogeneous Mixtures". In: *The*

Annals of Statistics 32.3 (2004), pp. 962–994. DOI: 10.1214/009053604000000265.

[6] Ronald A. Fisher. *Statistical Methods for Research Workers*. 4th ed. Edinburgh: Oliver and Boyd, 1932.

[7] Joshua Glasser and Brian Lindauer. "Bridging the Gap: A Pragmatic Approach to Generating Insider Threat Data". In: *2013 IEEE Security and Privacy Workshops*. San Francisco, CA, 2013, pp. 98–104. DOI: 10.1109/SPW.2013.37.

[8] Ivan Homoliak et al. "Insight Into Insiders and IT: A Survey of Insider Threat Taxonomies, Analysis, Modeling, and Countermeasures". In: *ACM Comput. Surv.* 52.2 (Apr. 2019). ISSN: 0360-0300. DOI: 10.1145/3303771. URL: https://doi.org/10.1145/3303771.

[9] Duc C. Le and A. Nur Zincir-Heywood. "Evaluating Insider Threat Detection Workflow Using Supervised and Unsupervised Learning". In: *2018 IEEE Security and Privacy Workshops (SPW)*. San Francisco, CA: IEEE, 2018, pp. 270–275. DOI: 10.1109/SPW.2018.00043.

[10] Duc C. Le and Nur Zincir-Heywood. "Anomaly Detection for Insider Threats Using Unsupervised Ensembles". In: *IEEE Transactions on Network and Service Management* 18.2 (2021), pp. 1152–1164.

[11] Brian Lindauer. *Insider Threat Test Dataset*. 2020. DOI: 10.1184/R1/12841247.v1.

[12] Lloyds Banking Group plc. *Annual Report and Accounts 2025*. Tech. rep. Published 13 February 2026; accessed 11 August 2026. Lloyds Banking Group plc, 2026.

[13] Douglas C. Montgomery. *Introduction to Statistical Quality Control*. 8th ed. Hoboken, NJ: Wiley, 2019.

[14] E. S. Page. "Continuous Inspection Schemes". In: *Biometrika* 41.1/2 (1954), pp. 100–115. DOI: 10.1093/biomet/41.1-2.100.

[15] Belinda Phipson and Gordon K. Smyth. "Permutation P-values Should Never Be Zero: Calculating Exact P-values When Permutations Are Randomly Drawn". In: *Statistical Applications in Genetics and Molecular Biology* 9 (2010). DOI: 10.2202/1544-6115.1585.

[16] Nils Reimers and Iryna Gurevych. "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks". In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Hong Kong, China: Association for Computational Linguistics, 2019, pp. 3982–3992. DOI: 10.18653/v1/D19-1410.

[17] Takaya Saito and Marc Rehmsmeier. "The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets". In: *PLOS ONE* 10.3 (Mar. 2015), pp. 1–21. DOI: 10.1371/journal.pone.0118432. URL: https://doi.org/10.1371/journal.pone.0118432.

[18] Mayu Sakurada and Takehisa Yairi. "Anomaly Detection Using Autoencoders with Nonlinear Dimensionality Reduction". In: *Proceedings of the MLSDA 2014 2nd Workshop on Machine Learning for Sensory Data Analysis*. 2014, pp. 4–11. DOI: 10.1145/2689746.2689747.

[19] Mei-Ling Shyu et al. "A Novel Anomaly Detection Scheme Based on Principal Component Classifier". In: *Proceedings of the IEEE Foundations and New Directions of Data Mining Workshop*. 2003, pp. 172–179.

[20] Zbyněk Šidák. "Rectangular Confidence Regions for the Means of Multivariate Normal Distributions". In: *Journal of the American Statistical Association* 62.318 (1967), pp. 626–633. DOI: 10.1080/01621459.1967.10482935.

[21] Leonard H. C. Tippett. *The Methods of Statistics*. London: Williams and Norgate, 1931.

[22] Wenhui Wang et al. "MINILM: deep self-attention distillation for task-agnostic compression of pre-trained transformers". In: NIPS '20. Vancouver, BC, Canada: Curran Associates Inc., 2020. ISBN: 9781713829546.

[23] Edwin B. Wilson. "Probable Inference, the Law of Succession, and Statistical Inference". In: *Journal of the American Statistical Association* 22.158 (1927), pp. 209–212. DOI: 10.1080/01621459.1927.10502953.

[24] Nong Ye and Qiang Chen. "An Anomaly Detection Technique Based on a Chi-Square Statistic for Detecting Intrusions into Information Systems". In: *Quality and Reliability Engineering International* 17.2 (2001), pp. 105–112.

[25] Dmitri V. Zaykin et al. "Truncated Product Method for Combining P-values". In: *Genetic Epidemiology* 22.2 (2002), pp. 170–185. DOI: 10.1002/gepi.0042.

Own Work Declaration

Generative AI (Claude, Anthropic; July–August 2026) was used in preparing this report as follows.

Writing: Used for word-level edits (all changes reviewed and accepted or rejected) and providing advice on where the report could be condensed to reach the word limit. Inputting some results into tables in LaTeX from code outputs.

Code: Debugging, stripping code that was no longer used, and editing the README note accompanying the code. Also used for editing some code docstrings (i.e.., automating shared inputs for many functions).

Checking: Verification of values reported in the text against code output after adjusting models to make sure my results hadn't changed, and identification of any clear errors in earlier drafts.

The research question, feature construction, choice of models and combination statistics, selection of operating points, and the conclusions drawn are my own.

Appendix

Word Count: 4993

Appendix A Email Volume Hourly Distribution

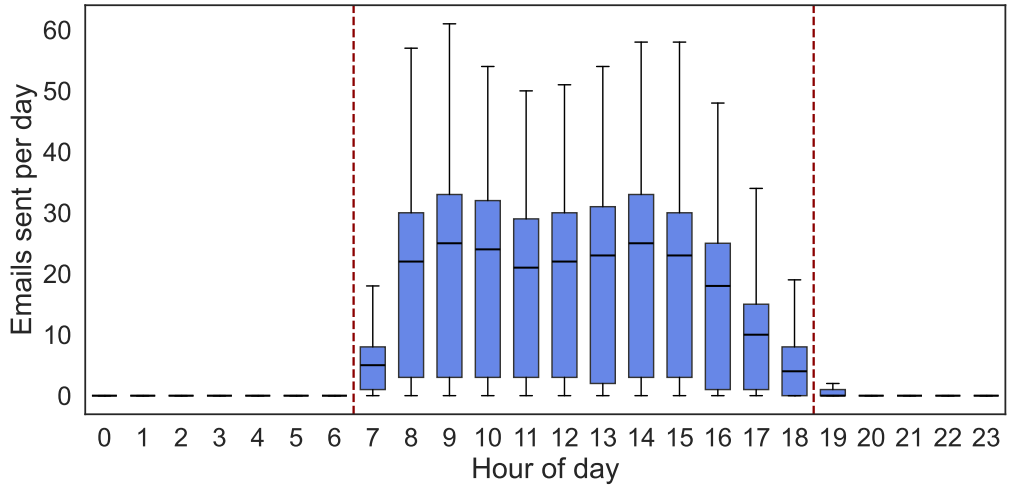


Figure 8: Distribution across days of the number of emails sent in each hour, over the full observation window. Boxes give the median and interquartile range; outliers are suppressed. Dashed red lines mark the 07:00 and 19:00 bounds within which a message is treated as sent during working hours (Section 2.2).

Appendix B Malicious Synthetic Scenarios

Table 9: The five malicious scenarios of CERT r6.2, quoted from the dataset documentation.

Scenario	Behaviour
1	User who did not previously use removable drives or work after hours begins logging in after hours, using a removable drive, and uploading data to wikileaks.org. Leaves the organization shortly thereafter.
2	User begins surfing job websites and soliciting employment from a competitor. Before leaving the company, they use a thumb drive (at markedly higher rates than their previous activity) to steal data.
3	System administrator becomes disgruntled. Downloads a keylogger and uses a thumb drive to transfer it to his supervisor's machine. The next day, he uses the collected keylogs to log in as his supervisor and send out an alarming mass email, causing panic in the organization. He leaves the organization immediately.
4	A user logs into another user's machine and searches for interesting files, emailing to their home email. This behavior occurs more and more frequently over a 3 month period.
5	A member of a group decimated by layoffs uploads documents to Dropbox, planning to use them for personal gain.

Appendix C Correlation Matrix of Features

Table 10: Pearson correlation between all eleven deviation features, computed after standardisation against each sender's own baseline (Section 2.2). Lower triangle only; the upper is its transpose. Column indices correspond to the numbered rows.

	1	2	3	4	5	6	7	8	9	10
1. `log_size`										
2. `n_recipients`	0.00									
3. `is_external`	0.00	−0.08								
4. `n_attachments`	**0.72**	0.00	0.00							
5. `is_weekend`	0.00	0.00	−0.01	0.00						
6. `is_after_hours`	0.00	0.01	0.00	0.00	0.01					
7. `log_volume_7d`	0.01	−0.01	0.01	0.01	0.01	0.00				
8. `external_sender`	0.00	−0.16	**0.89**	0.00	−0.01	0.00	0.00			
9. `contact_surprise`	0.00	0.07	0.55	0.00	−0.01	0.00	0.00	0.55		
10. `frac_novel_recipients`	0.00	0.01	0.04	0.00	0.00	−0.01	−0.02	0.03	0.45	
11. `semantic`	0.00	0.14	0.23	−0.01	0.00	0.00	−0.01	0.26	0.20	0.05

Appendix D Ranking Model Performance Across All Cutoffs

Table 11: Model performance summarised over the whole ranking rather than at a single cutoff. PR-AUC is average precision; against the evaluation set's base rate of 0.1% (Section 2.1), the ensemble's 0.0156 represents roughly a sixteen-fold lift. AP@10% is average precision restricted to the leading 10% of the ranking and normalised to that range, matching the screening budget used in Table 3.

Model	Regime	PR-AUC	ROC-AUC	AP@10%
Z-score	—	0.0133	0.7665	0.0039
PCA	Novelty	0.0111	0.9000	0.0107
Autoencoder	Novelty	0.0156	0.9059	0.0139
Ensemble	Novelty	0.0158	0.9099	0.0139
PCA	All data	0.0110	0.8996	0.0106
Autoencoder	All data	0.0157	0.9075	0.0140
Ensemble	All data	0.0156	0.9106	0.0139

Table 11 confirms the ordering of Section 4.1 without reference to any cutoff, and reproduces the regime finding of Section 4.3: no model differs by more than 0.0002 on PR-AUC or AP@10%, or 0.002 on ROC-AUC, between regimes.

Appendix E Feature Importance by Scenario

Table 12: Recall lost when each feature is shuffled, in percentage points (pp), measured separately within each scenario's own positives. Same shuffles, same fitted models and same 5% cutoff as Table 5; only the denominator changes. Baseline recalls are 100.0% (Scenario 2, 13 emails), 6.7% (Scenario 3, 15 emails), 68.6% (Scenario 4, 70 emails) and 63.3% overall. One email is worth 7.7pp in Scenario 2, 6.7pp in Scenario 3 and 1.4pp in Scenario 4, so the scenario columns are far coarser than the overall column; the standard deviation is across five shuffles and is reported for the overall column only.

Feature	S2	S3	S4	All	sd
`log_volume_7d`	0.0	−4.0	**54.6**	38.4	2.3
`external_sender`	35.4	−4.0	33.1	27.8	4.0
`is_external`	20.0	−5.3	26.0	20.4	2.2
`contact_surprise`	**64.6**	0.0	12.6	17.6	1.3
`n_attachments`	6.2	−8.0	16.9	11.6	1.4
`frac_novel_recipients`	16.9	−1.3	11.1	10.0	0.9
`log_size`	6.2	−5.3	10.3	7.4	1.5
`n_recipients`	10.8	0.0	7.4	6.7	0.9
`semantic`	0.0	0.0	4.0	2.9	0.9
`is_after_hours`	0.0	6.7	−1.7	−0.2	0.5
`is_weekend`	0.0	0.0	−3.4	−2.5	0.9

Table 12 separates two detection mechanisms that the corpus-wide ordering of Table 5 combines. Scenario 2 is recovered almost entirely through the recipient it is addressed to: `contact_surprise` alone costs 64.6 of its 100 points of recall when shuffled, and `log_volume_7d` costs none of it. Scenario 4 is recovered chiefly through sending rate: `log_volume_7d` costs 54.6 points against a 68.6% baseline, some four fifths of that scenario's detection.

Table 5 therefore reports a weighted average in which Scenario 4, contributing 70 of the 98 positives, dominates. This is why `log_volume_7d` leads the corpus-wide ranking despite being worth nothing to Scenario 2, and why `contact_surprise` places only fourth despite being the single most important feature for the scenario its construction targets.

Scenario 3's results should not be used for inference. Its baseline recall is a single email of fifteen, so its per-shuffle granularity is 6.7 points; the +6.7 against `is_after_hours` is one email recovered by chance under permutation, not a feature the model relies on inversely.

Appendix F Weekly Detection Record for the Accounts of Interest

Table 13: Every in-window account-week for the four accounts of interest. $\tilde{z}_F$ and $\tilde{z}_T$ are the centred Fisher and Tippett statistics (Section 6.2), ranked as R_F and R_T among the pool of accounts assessed that week; C is the cumulative sum of Equation 7, and an investigation opens where it exceeds $h = 6$. R_{avg} is the rank the account would have held under a single averaged statistic $(\tilde{z}_F + \tilde{z}_T)/2$. Table 7 reports only the weeks reaching the leading three, so this is the record behind the rank claims of Section 6.4. Bold marks the two CUSUM crossings and the week recovered by the Tippett stream. CMP2946's thirteenth malicious email (16 March 2011) falls outside these weeks.

Account	Week ending	Em.	Mal.	Pool	$\tilde{z}_F$	R_F	C	$\tilde{z}_T$	R_T	R_{avg}
CDE1846	27 Feb 2011	14	3	83	0.09	20	0.00	0.62	15	18
	06 Mar 2011	10	0	77	−0.11	29	0.00	0.13	27	25
	13 Mar 2011	12	0	78	0.22	17	0.00	2.05	5	9
	20 Mar 2011	19	3	78	0.38	12	0.00	−0.10	31	25
	27 Mar 2011	31	13	78	8.25	1	**7.75**	1.92	4	1
	03 Apr 2011	15	6	78	2.93	2	2.43	2.11	7	2
	10 Apr 2011	21	6	78	0.71	9	2.64	1.58	4	5
	17 Apr 2011	40	22	82	8.47	1	**10.61**	3.33	2	1
	24 Apr 2011	14	0	80	0.35	16	0.00	2.76	2	4
	01 May 2011	33	17	77	5.44	1	4.94	0.65	12	2
CMP2946	13 Feb 2011	29	2	81	−0.27	38	0.00	0.43	19	28
	20 Feb 2011	36	4	85	−0.36	47	0.00	0.22	25	34
	27 Feb 2011	34	6	83	5.08	1	4.58	1.81	5	1
	06 Mar 2011	27	0	77	−0.19	42	3.90	1.64	6	8
HIS1706	15 Aug 2010	32	3	83	−0.22	53	0.00	**4.29**	**1**	3
PLJ1771	15 Aug 2010	22	2	83	−0.15	41	0.00	−0.06	37	40

Table 13 carries the evidence for Section 6.4. The two accounts not recovered are visible here in a way Table 7 cannot show. `CMP2946` ranks 38th and 47th in the two weeks preceding its single first-ranked week, so no evidence accumulates beforehand and C reaches only 4.58. `PLJ1771` ranks 41st and 37th in its single in-window week, mid-pool on both statistics: its two malicious emails are 97 minutes apart on an account whose week is otherwise unremarkable, and no email-based statistic separates it.

The R_{avg} column shows what combining the two statistics would cost. Averaging demotes any account-week that is extreme on one statistic alone, in both directions: the weeks ending 13 March and 24 April place fifth and second on Tippett but ninth and fourth on the average, while the week ending 20 March places twelfth on Fisher and twenty-fifth on the average. Only weeks extreme on both survive the combination unchanged, which is precisely the pattern neither statistic was constructed to detect.

One account pays for this. `HIS1706` is first on Tippett and third under averaging, and the point-anomaly stream reads one message a week; its Fisher rank of 53 is enough to displace it from a queue of one. Third of 83 is not, in itself, a poor placing, so the streams are not merely differently ordered but differently budgeted, and averaging discards the ordering the smaller budget depends on. `CDE1846`, by contrast, ranks first under averaging in both of the weeks its cumulative sum crosses h, so the split queue costs nothing on the account it recovers.